NIZWA COLLEGE OF TECHNOLOGY
DEPARTMENT OF INFORMATION TECHNOLOGY

SET-A

الكلية التقنية بنزوى
Nizwa College of Technology

Final Examination

Semester 1, Academic Year 2018-2019

Course Code: MATH1200 Course Title: CALCULUS - 1

Date: 16-12-2018 **Time:** 9 AM – 11 AM

Max Marks	50	Time Allocation	2 Hours
Student ID		Student Name	
Section		Level	DIPLOMA

Instructions for students

1. Exam paper consists of 8 pages including cover page. If any page is missing, inform the invigilator immediately.
2. Answer the questions in the space provided by using **blue or black pen** only.
3. Show the steps for the answers clearly.
4. Calculator is allowed.
5. Write the correct units, if any.

Question Type	Total Marks	Marks Obtained
PART A - Multiple Choice Questions	$10 \times 1 = 10$	
PART B - Short Answer Questions	$7 \times 4 = 28$	
PART C - Long Answer Questions	$2 \times 6 = 12$	
Total Marks	**50**	

Question Setter	Moderator	HOS
Dr. MOAZZAM ARIF Moazzam Arif	09.12.2018	For 10/12/2018 (Manithe Raju)

1. (ANITHA)
2. (SUMANTHI)

PART A – MULTIPLE CHOICE QUESTIONS

Answer all questions. Each question carries 1 mark.

A1. The absolute maximum value of $f(x) = 3 - x^2$ on the interval $[-9, 9]$ is _____.

(a) $\sqrt{2}$ (b) 2 (c) 3 (d) $\sqrt{3}$

A2. If $y = 2x \ln x$, then $y' =$ _____.

(a) $1 + \ln x$ (b) $1 - \ln x$ (c) $2(1 + \ln x)$ (d) $2(1 - \ln x)$

A3. $\lim_{x \to 2} \dfrac{x^2 - 4}{x - 2} =$ _____.

(a) 4 (b) 2 (c) 0 (d) -2

A4. The derivative of a constant function is _____.

(a) 0 (b) 1 (c) -1 (d) ∞

A5. The height of an object moving vertically is given by $s = -t^2 + 6t + 11$ with s in feet and t in seconds. The time when the height will be maximum is _____.

(a) 6 secs (b) 5 secs (c) 4 secs (d) 3 secs

A6. $\int 3e^x dx =$ _____.

(a) $e^x + c$ (b) $5e^x + c$ (c) $3e^x + c$ (d) 3

A7. If $f'(x) = x - \cos^{-1} x$, then $f''(x) =$ _____.

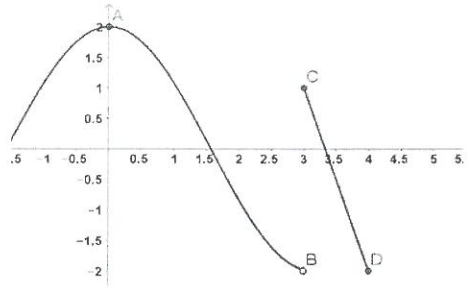
(a) $1 - \dfrac{1}{\sqrt{1+x^2}}$ (b) $1 + \dfrac{1}{\sqrt{1+x^2}}$ (c) $1 - \dfrac{1}{\sqrt{1-x^2}}$ (d) $1 + \dfrac{1}{\sqrt{1-x^2}}$

A8. $\int \cos 2x \, dx =$ _____.

(a) $\dfrac{\sin 2x}{2} + c$ (b) $\dfrac{-\sin 2x}{2} + c$ (c) $\dfrac{\cos 2x}{2} + c$ (d) $\dfrac{-\cos 2x}{2} + c$

A9. If $xy = x - 11$, then $\dfrac{dy}{dx} =$ _____.

(a) $\dfrac{-y}{x}$ (b) $\dfrac{1-y}{x}$ (c) $\dfrac{1+y}{x}$ (d) $\dfrac{y}{x}$



A10. From the figure, $\lim_{x \to 3^-} f(x)$ is _____.

(a) -1 (b) -2 (c) 0 (d) 1

PART B – SHORT ANSWER QUESTIONS

Answer all questions. Each question carries 4 marks.

B1. Use second derivative test to find the point(s) where the given function $f(x) = 3x^3 - 6x^2 + 5$ has extreme values.

$f'(x) = 9x^2 - 12x$ (0.5)

$f'(x) = 0 \Rightarrow 9x^2 - 12x = 0$

$\Rightarrow 3x(3x-4) = 0$

$\Rightarrow x = 0, \frac{4}{3}$ (1)

$f''(x) = 18x - 12$ (0.5)

$f''(x)_{(0)} = -12 < 0$ $f(x)$ has maximum at $x=0$ (1)

$f''(x)_{(\frac{4}{3})} = 18 \cdot \frac{4}{3} - 12 = 24 - 12 = 12 > 0$ so, $f(x)$ has minimum at $x = \frac{4}{3}$. (1)

B2. For what value(s) of k, the function defined by $f(x) = \begin{cases} kx^2 - 4, & x < 1 \\ 3x + 2, & x \geq 1 \end{cases}$ is continuous at $x = 1$.

We have $LHL = RHL = f(1)$

$LHL = \lim_{x \to 1^-} f(x) = \lim_{x \to 1} kx^2 - 4 = k - 4$ (1)

$RHL = \lim_{x \to 1^+} f(x) = \lim_{x \to 1} 3x + 2 = 3 \times 1 + 2 = 5$ (1)

$LHL = RHL$ (1)

$k - 4 = 5$

$\boxed{k = 9}$ (1)

B3. Find the derivative of $r = \left(\dfrac{\sin\theta}{1+\cos\theta}\right)^3$ with respect to θ.

$$\frac{dr}{d\theta} = 3\left(\frac{\sin\theta}{1+\cos\theta}\right)^2 \cdot \frac{d}{d\theta}\left(\frac{\sin\theta}{1+\cos\theta}\right) \quad (2)$$

$$= 3\,\frac{\sin^2\theta}{(1+\cos\theta)^2} \cdot \frac{(1+\cos\theta)\cos\theta - \sin\theta(-\sin\theta)}{(1+\cos\theta)^2} \quad (1)$$

$$= 3\,\frac{\sin^2\theta}{(1+\cos\theta)^4} \cdot (1+\cos\theta)$$

$$\therefore \frac{dr}{d\theta} = \frac{3\sin^2\theta}{(1+\cos\theta)^3} \quad (1)$$

B4. For the given curve $y = x^3 - 12x$,

(a) find all points on the graph where the tangent line is parallel to the x-axis. [2]

(b) Also find the equation of horizontal tangents at all these points. [2]

a) $\dfrac{dy}{dx} = 0$, for parallel tangent

$3x^2 - 12 = 0$

$3(x^2-4) = 0$

$x = \pm 2$ (1)

$x = 2$, then $y = -16$

$x = -2$, then $y = 16$

$\therefore$ points are $(2,-16), (-2,16)$ (1)

b) For $(2,-16)$

eqn. of tangent : $y - y_1 = m(x - x_1)$

$y + 16 = 0$

$y = -16$ (1)

For $(-2,16)$: $y - y_1 = m(x - x_1)$

$y - 16 = 0$

$y = 16$ (1)

B5. Verify Rolle's Theorem for the function $f(x) = x^2 - 17x + 16$ on the interval $[1,16]$

i) $f(x)$ is defined and continuous in closed interval $[1,16]$ (0.5)

ii) $f(x)$ is differentiable in open interval $(1,16)$ (0.5)

iii) $f(1) = 1^2 - 17\times1 + 16 = 0$, $f(16) = 16^2 - 17\times16 + 16$
$= 256 - 272 + 16 = 0$

so, $f(1) = f(16)$ (1)

$f'(c) = 0$, $2c - 17 = 0$

$c = \frac{17}{2} = 8.5$ (1)

$c \in (1,16)$ and hence Rolle's Theorem Verified. (1)

B6. Evaluate the integrals

(a) $\int_0^{\frac{\pi}{6}} 3\sec^2 x\, dx$ ~~[4]~~ [2]

$= 3\int_0^{\pi/6} \sec^2 x\, dx$

$= 3\left[\tan x\right]_0^{\pi/6}$ (1)

$= 3\left[\tan\frac{\pi}{6} - \tan 0\right]$

$= 3\left[\frac{1}{\sqrt{3}} - 0\right]$

$= \frac{3}{\sqrt{3}} = \sqrt{3}$ (1)

(b) $\int\left(\frac{1}{x^3}+3\right)dx$ [2]

$= \int x^{-3}dx + 3\int dx + C$ (1)

$= \frac{x^{-3+1}}{-3+1} + 3x + C$

$= -\frac{1}{2x^2} + 3x + C$ (1)

B7. If $x = 3t^2 + 2$ and $y = 2t\tan t$, then find $\frac{dy}{dx}$ at $t=\pi/4$

$\frac{dx}{dt} = \frac{d}{dt}(3t^2+2)$

$= 6t$ (0.5)

$\frac{dy}{dt} = \frac{d}{dt}(2t\tan t)$

$= 2[t\sec^2 t + \tan t]$ (1)

$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$ (0.5)

$= \frac{\cancel{2}[t\sec^2 t + \tan t]}{3\cancel{6}t}$

$\frac{dy}{dx} = \frac{t\sec^2 t + \tan t}{3t}$ (1)

$\left.\frac{dy}{dx}\right]_{t=\pi/4} = \frac{\frac{\pi}{4}\cdot 2 + 1}{3\cdot\pi/4} = \frac{\frac{\pi}{2}+1}{\frac{3\pi}{4}}$ (1)

PART C – LONG ANSWER QUESTIONS

Each question carries 6 marks.

C1. Given $y = 5x^3 + 15x^2 - 45x + 10$. Find the intervals in which the function is increasing/decreasing, concave up/concave down, then identify the local extreme values and inflection points.

$y' = 15x^2 + 30x - 45 = 0$ for critical pts.

$15(x^2 + 2x - 3) = 0$

$x^2 + 3x - x - 3 = 0$

$(x+3)(x-1) = 0$

$x = -3, 1$ → (1)

intervals	$(-\infty, -3)$	$(-3, 1)$	$(1, \infty)$
Sign of f'	+	−	+
behaviour of f	increasing	decreasing	increasing

(1.5)

$x = -3$ is a pt. of local maximum and $x = 1$ is a pt. of local minimum. (1)

maximum value at $x = -3$ is $5(-3)^3 + 15(-3)^2 - 45 \times -3 + 10$

$= 5(-27) + (15 \times 9) + 135 + 10$

$= 145$

minimum value $= 5(1)^3 + 15(1)^2 - 45 \times 1 + 10 = -15$] (1)

Now, $f''(x) = 0 \Rightarrow 30x + 30 = 0$

$\Rightarrow \boxed{x = -1}$

intervals	$(-\infty, -1)$	$(-1, \infty)$
Sign of f''	−	+
Concavity	concave down	Concave up

(1)

$x = -1$ is inflection point. (0.5)

C2. The position of a moving particle is given as $s = -t^2 + 6t + 2, 0 \le t \le 1$ with s in meters and t in seconds. Find at what time the average velocity is equal to the instantaneous velocity. Also find $\frac{d^2s}{dt^2}$ at $t = 1$

$\Delta s = f(1) - f(0)$

$= (-1+6+2) - 2 = 5$ (1)

$\Delta t = 1 - 0 = 1$

average velocity $= \frac{\Delta s}{\Delta t} = \frac{5}{1} = 5$ m/s (1)

instantaneous velocity, $\frac{ds}{dt} = -2t+6$ (1)

$\frac{\Delta s}{\Delta t} = \frac{ds}{dt}$

average velocity = instantaneous velocity (1)

$5 = -2t+6$

$-1 = -2t$

$\Rightarrow t = 0.5$ seconds (1)

$\frac{ds}{dt} = -2t+6$

$\frac{d^2s}{dt^2} = -2$

$\left[\frac{d^2s}{dt^2}\right]_{t=1} = -2$ (1)

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Dr. MOAZZAM ARIF Moazzam Arif	09.12.2018	For 10/12/2018

1. (ANITHA)
2. (SWANTHI)

(Manitha Rijo)

PART A – MULTIPLE CHOICE QUESTIONS

Answer all questions. Each question carries 1 mark.

A1. The absolute maximum value of $f(x) = 2 - x^2$ on the interval $[-4, 4]$ is _____.

(a) $\sqrt{2}$ (b) 2 (c) 3 (d) $\sqrt{3}$

A2. If $y = x \ln x$, then $y' =$ _____.

(a) $1 + \ln x$ (b) $1 - \ln x$ (c) $-\ln x$ (d) $\ln x$

A3. $\lim_{x \to 1} \dfrac{x^2 - 1}{x - 1} =$ _____.

(a) -2 (b) 0 (c) 1 (d) 2

A4. The derivative of a constant function is _____.

(a) $-\infty$ (b) -2 (c) -1 (d) 0

A5. The height of an object moving vertically is given by $s = -t^2 + 4t + 11$ with s in feet and t in seconds. The time when the height will be maximum is _____.

(a) 4 secs (b) 3 secs (c) 2 secs (d) 1 sec

A6. $\int 5e^x dx =$ _____.

(a) $5e^x + c$ (b) $5x + c$ (c) $5xe^x + c$ (d) 5

A7. If $f'(x) = x - \sin^{-1} x$, then $f''(x) =$ _____.

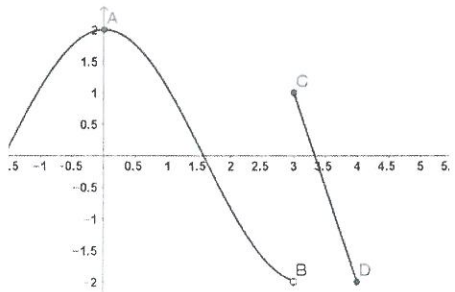
(a) $1 - \dfrac{1}{\sqrt{1+x^2}}$ (b) $1 + \dfrac{1}{\sqrt{1-x^2}}$ (c) $1 - \dfrac{1}{\sqrt{1-x^2}}$ (d) $1 + \dfrac{1}{\sqrt{1+x^2}}$

A8. $\int \cos 4x \, dx =$ _____.

(a) $\dfrac{-\sin 4x}{4} + c$ (b) $\dfrac{\sin 4x}{4} + c$ (c) $\dfrac{\cos 4x}{4} + c$ (d) $\dfrac{-\cos 4x}{4} + c$

A9. If $xy = x - 12$, then $\dfrac{dy}{dx} =$ _____.

(a) $\dfrac{-y}{x}$ (b) $\dfrac{1+y}{x}$ (c) $\dfrac{1-y}{x}$ (d) $\dfrac{y}{x}$



A10. From the figure, $\lim_{x \to 3^+} f(x)$ is _____.

(a) -2 (b) -1 (c) 0 (d) 1

PART B – SHORT ANSWER QUESTIONS

Answer all questions. Each question carries 4 marks.

B1. Use second derivative test to find the point(s) where the given function $f(x) = 2x^3 - 5x^2 + 7$ has extreme values.

$f'(x) = 6x^2 - 10x$ (0.5)

$f'(x) = 0 \Rightarrow 2x(3x-10) = 0$

$\Rightarrow x = 0,\ x = \frac{10}{3}$ (1)

$f''(x) = 12x - 10$ (0.5)

$f''(x)$ at $x=0$ $= -10 < 0$, $f(x)$ has maxm^m at $x=0$ (1)

$f''(x)$ at $x = \frac{10}{3}$ $= 12 \times \frac{10}{3} - 10 = 30 > 0$, $f(x)$ has minimum at $x = \frac{10}{3}$ (1)

B2. For what value(s) of m, the function defined by $f(x) = \begin{cases} mx^2 - 4, & x < 1 \\ 5x + 2, & x \geq 1 \end{cases}$ is continuous at $x = 1$.

LHL = RHL = $f(1)$

LHL $= \lim_{x \to 1^-} f(x) = \lim_{x \to 1} mx^2 - 4 = m - 4$ (1)

RHL $= \lim_{x \to 1^+} f(x) = \lim_{x \to 1} 5x + 2 = 5 \times 1 + 2 = 7$ (1)

LHL = RHL $\Rightarrow m - 4 = 7$

$\Rightarrow m = 11$ (1)

(1)

B3. Find the derivative of $r = \left(\dfrac{\sin\theta}{1+\cos\theta}\right)^5$ with respect to θ.

$$\frac{dr}{d\theta} = 5\left(\frac{\sin\theta}{1+\cos\theta}\right)^4 \cdot \frac{d}{d\theta}\left(\frac{\sin\theta}{1+\cos\theta}\right) \quad (2)$$

$$= \frac{5\sin^4\theta}{(1+\cos\theta)^4} \cdot \frac{(1+\cos\theta)\cos\theta - \sin\theta(-\sin\theta)}{(1+\cos\theta)^2} \quad (1)$$

$$= \frac{5\sin^4\theta}{(1+\cos\theta)^4} \cdot \frac{(1+\cos\theta)}{(1+\cos\theta)^2}$$

$$\frac{dr}{d\theta} = \frac{5\sin^4\theta}{(1+\cos\theta)^5} \quad (1)$$

B4. For the given curve $y = x^3 - 3x$,

(a) find all the points on the graph where the tangent line is parallel to the x-axis. [2]

(b) Also find the equation of horizontal tangents at all these points. [2]

a) $\frac{dy}{dx} = 0$, for parallel tangents

$3x^2 - 3 = 0$

$3(x^2-1) = 0$

$x = \pm 1$ (1)

For $x = 1$, $y = -2$

$x = -1$, $y = 2$

$\therefore$ pts are $(-1,2)$, $(1,-2)$ (1)

b) For $(-1,2)$, horizontal tangent:

$y - y_1 = m(x - x_1)$

$(y-2) = 0(x+1)$

$y - 2 = 0$, $\boxed{y = 2}$ (1)

For $(1,-2)$,

$y - y_1 = m(x - x_1)$

$y + 2 = 0$

$\boxed{y = -2}$ (1)

B5. Verify Rolle's Theorem for the function $f(x) = x^2 - 20x + 19$ on the interval $[1,19]$

i) $f(x)$ is defined and continuous in $[1,19]$ (0.5)

ii) $f(x)$ is differentiable in $(1,19)$ (0.5)

iii) $f(1) = 1^2 - 20\times1 + 19 = 0$

$f(19) = 19^2 - 20\times19 + 19 = 0$

$f(1) = f(19)$ (1)

then, $f'(c) = 0$

$\Rightarrow 2c - 20 = 0$

$\Rightarrow c = 10$ (1)

$c \in (1,19)$

Hence Rolle's Theorem is verified. (1)

B6. Evaluate the integrals

(a) $\int_0^{\frac{\pi}{3}} 2\sec^2 x\, dx$ [2]

$2\int_0^{\pi/3} \sec^2 x\, dx$

$2\left[\tan x\right]_0^{\pi/3}$ (1)

$2\left[\tan \pi/3 - \tan 0\right]$

$2(\sqrt{3} - 0)$

$= 2\sqrt{3}$ (1)

(b) $\int\left(\frac{1}{x^2}+2\right)dx$ [2]

$= \int x^{-2}dx + 2\int dx + \cancel{c}$ (1)

$= \frac{x^{-1}}{-1} + 2x + c$

$= -\frac{1}{x} + 2x + c$ (1)

B7. If $x = 2t^3 + 1$ and $y = t^2 \tan t$, then find $\frac{dy}{dx}$ at $t = \frac{\pi}{4}$

$\frac{dx}{dt} = 6t^2$ (0.5)

$\frac{dy}{dt} = \frac{d(t^2 \tan t)}{dt}$

$= t^2 \sec^2 t + 2t \tan t$ (1)

$\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$ (0.5)

$= \frac{t^2 \sec^2 t + 2t \tan t}{6t^2}$

$= \frac{\cancel{t}(t \sec^2 t + 2\tan t)}{6t^{\cancel{2}}}$

$= \frac{t \sec^2 t + 2\tan t}{6t}$ (1)

$\left.\frac{dy}{dx}\right]_{t=\pi/4} = \frac{\frac{\pi}{4}\cdot 2 + 2}{6\cdot\frac{\pi}{4}} = \frac{\frac{\pi}{2}+2}{3\pi/4}$ (1)

PART C – LONG ANSWER QUESTIONS

Each question carries 6 marks.

C1. Given $y = 4x^3 + 12x^2 - 36x + 8$. Find the intervals in which the function is increasing/decreasing, concave up/concave down, then identify the local extreme values and inflection points.

$y' = 12x^2 + 24x - 36$

For critical pts, $y' = 0$

$12(x^2 + 2x - 3) = 0$

$x = -3, 1$ → (1)

intervals	$(-\infty, -3)$	$(-3, 1)$	$(1, \infty)$	
sign of f'	+	−	+	(1.5)
behaviour of f	increasing	decreasing	increasing	

$x = -3$ is a pt. of local maximum and $x = 1$ is a pt. of local minimum. (1)

maximum value $= 4(-3)^3 + 12(-3)^2 - 36(-3) + 8$

$= 4(-27) + 12 \times 9 + 108 + 8$

$= 116$ (1)

minimum value $= 4(1)^3 + 12 \times 1^2 - 36 \times 1 + 8 = -12$

Now, $f''(x) = 0 \Rightarrow 24x + 24 = 0$

$\Rightarrow x = -1$

intervals	$(-\infty, -1)$	$(-1, \infty)$	
sign of f''	−	+	(1)
concavity	Concave down	Concave up	

$x = -1$ is an inflection point. (0.5)

C2. The position of a moving particle is given as $s = -t^2 + 8t + 2, 0 \le t \le 1$ with s in meters and t in seconds. Find at what time the average velocity is equal to the instantaneous velocity. Also find $\frac{d^2s}{dt^2}$ at $t = 1$

$\Delta s = f(1) - f(0)$

$= (-1+8+2) - 2$

$= 7$ (1)

$\Delta t = 1 - 0 = 1$

average velocity $= \frac{\Delta s}{\Delta t} = \frac{7}{1} = 7$ m/s (1)

when $\frac{\Delta s}{\Delta t} = \frac{ds}{dt}$ (1) instantaneous velocity: $\frac{ds}{dt} = -2t+8$ (1)

$7 = -2t + 8$

$+1 = +2t$

$t = 0.5$ sec. (1)

Now, $\frac{ds}{dt} = -2t + 8$

$\frac{d^2s}{dt^2} = -2$

$\left.\frac{d^2s}{dt^2}\right]_{t=1} = -2$ (1)